

IAC

Monday, September 21, 2026 10:19 AM

Infrastructure as code.

What you need IAC.

1. Reusability
2. Eliminating human error.
3. Simplicity

Walmart

They want to migrate there customer facing application to the cloud.

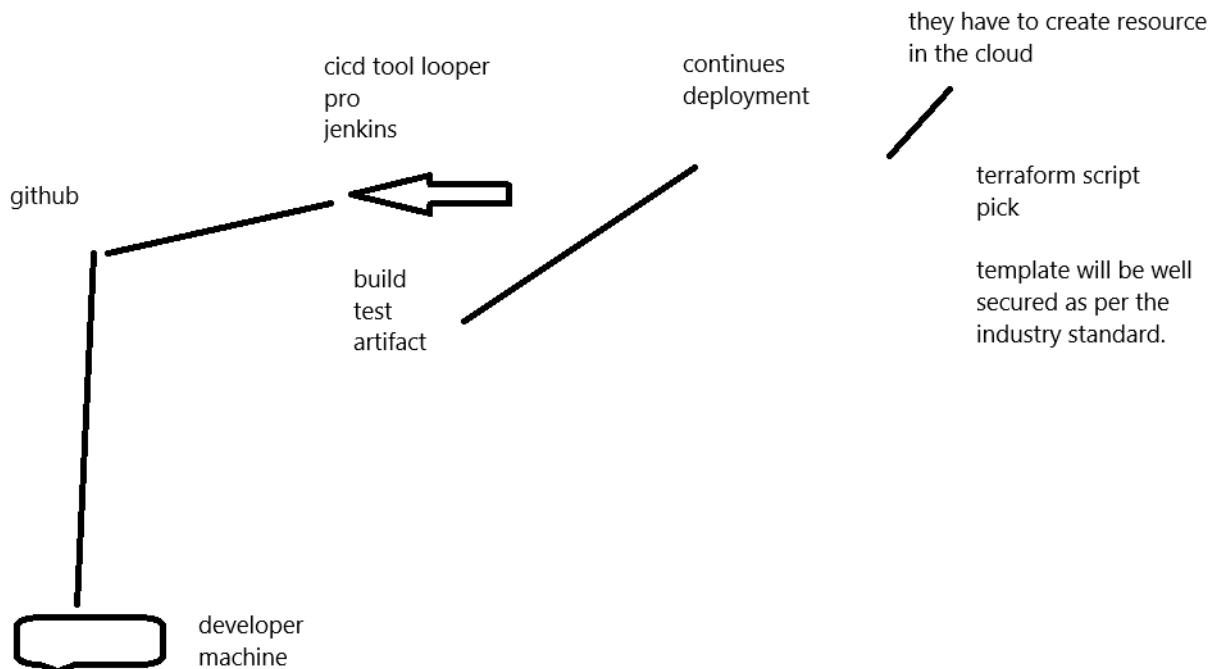
Azure where they will deploy application

GCP: data analytics

They have more than 10k application.

When they deploy they deploying in two different region inside a country.

As it is retails frequent always required.



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IAC :

Configuration management: where it will be applied in existing resource ansible chef puppet

Provision tool: terraform. Which is used to build an infra in the cloud

Aws: cft

Azure: arm template

Gcp: json terraform.

1. Backup
2. Dr location.
3. Azure is good for application deployment HIPAA
4. Gcp: data analytics: PCI DSS

One single language to manage all your infra.

If you know how cloud behave inside your cloud how the resources are being created using the cloud console

#same way I need to find out the code from registry.terraform.io and arrange it.

I want to create a virtual machine in azure cloud.

1. Subscription.

We build the resources in the cloud when we need it and when we need it we also need to destroy.

What is the problem

If I delete the resource group everything will be deleted inside it

In aws. I have created rds database. They have created the endpoint

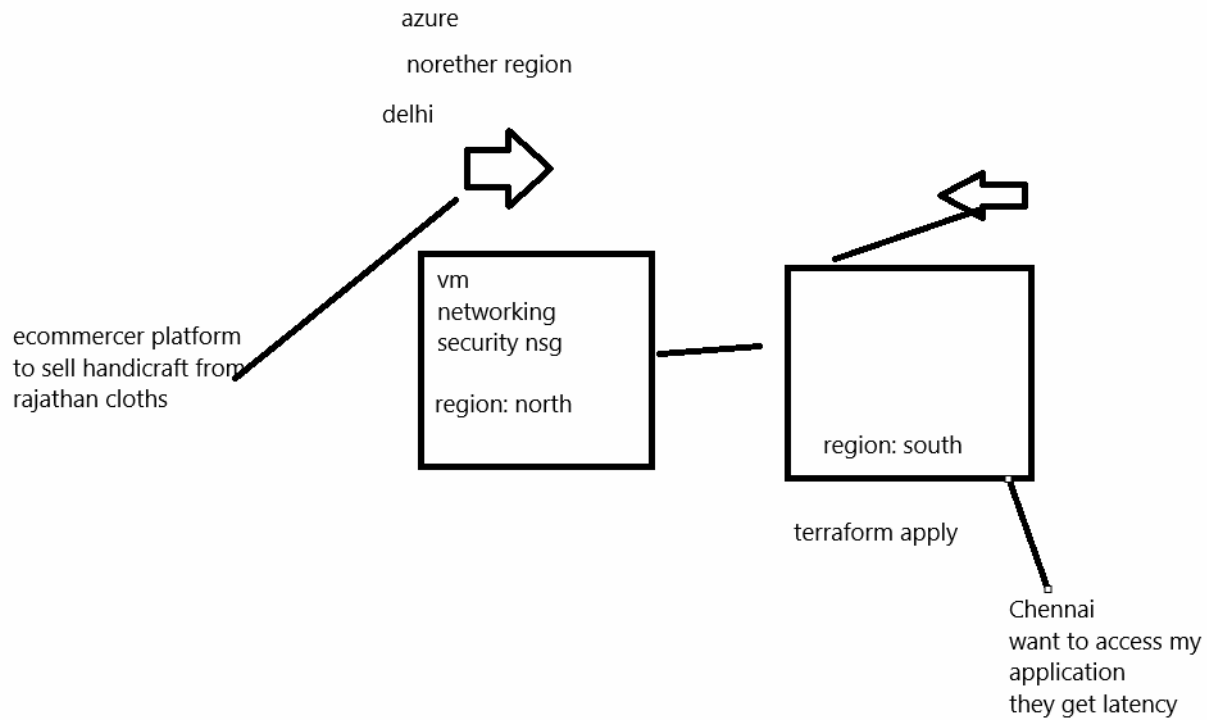
I delete the rds database. This end point is not deleted. 600\$. Aws reverse it.

Azure. I deleted all the resource devops resource is not deleted a month to reverse the charges.

I destroy means everything should go away.

It create resource store the information inside a file

Destroy means it read that file and destroy everything.



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Terraform installation

Monday, September 21, 2026 11:34 AM

To work with terraform you need a workstation.

Inside the workstation you will have

1. Terraform package
2. Cloud cli: azure
3. Git client
4. Vs code. This vs code editor will help you to write your code faster give you suggestion and you can easily capture the error.

#we will install a windows machine in azure cloud

#we will take the rdp of the windows machine

@inside the windows vm we will install all this.

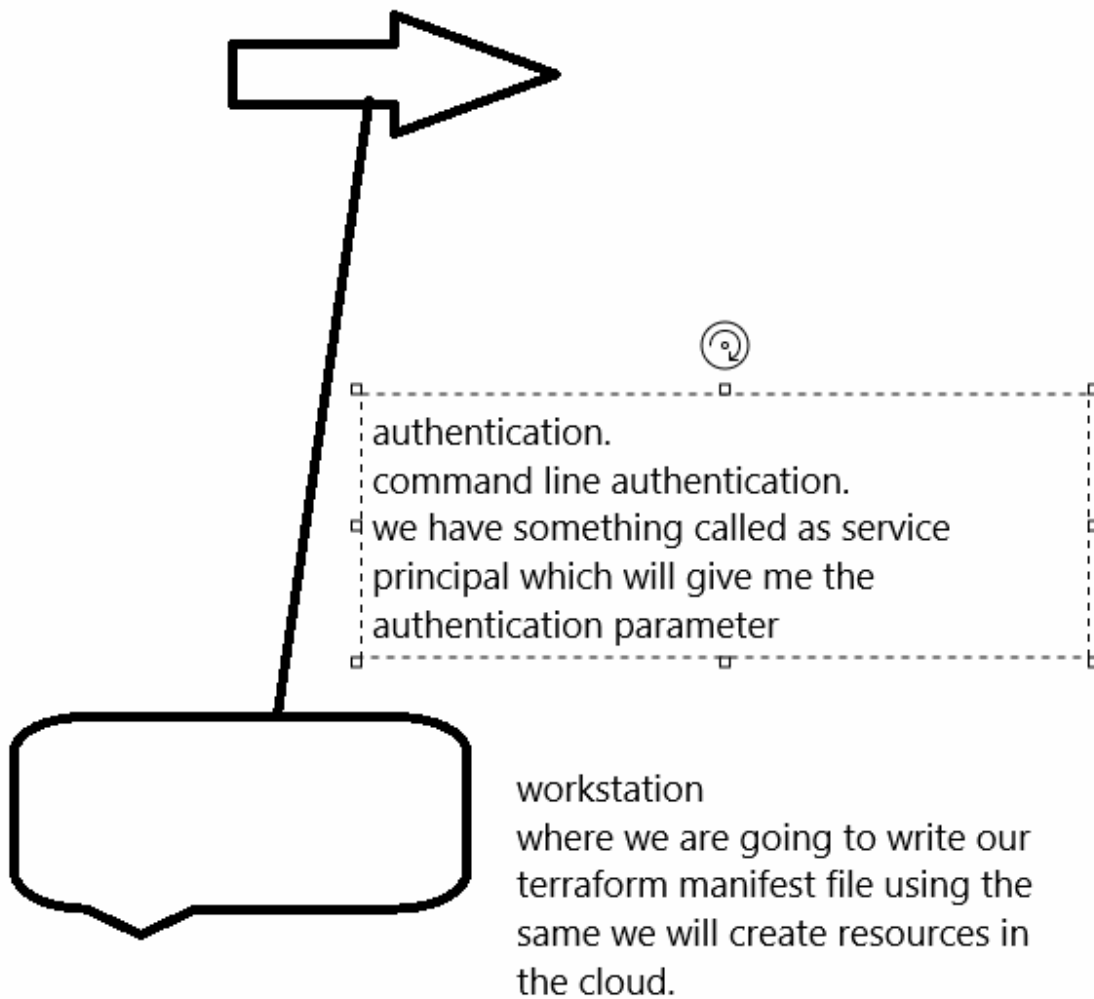
Disconnect vpn

Disconnet ghlobal protect

Disconnect proxy cap.

Choco is a package manager similar to yum or apt.

azure cloud



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Terraform workflow

Monday, September 21, 2026 2:05 PM

When you create any script

How to run the same

Python `python nameofthefile.py`

Bash `nameoffile.sh`

Every terraform command will always run on current directory

We create a directory : inside the directory we will have manifest file.

Code to create to create resource `vpnet subnet vm storage`.

Inside the current directory we will run first command

Terraform init: inside the current directory it will download the cloud service provider. Create a `.terraform` folder and download the plugin which will be in exe format.

Terraform plan: read all your manifest file. Compare it with cloud service provider. And give you're a dry run. Skelton what is going to create.

Terraform apply: again it will read all your manifest file. Compare it with the cloud service provider.

Current state to desired state. Yes it will convert it to desired state. Once converted into desired state it will store the state information inside a file `terraform.tfstate`

Terraform destroy: destroy it. Which file it is going to read. Log or state file.

Terraform configuration

Monday, September 21, 2026 2:20 PM

#using terraform what we are creating.

Resources - block resources

I am create vnet as a resource

Argument

Vnet name and

Cidr block.

Nameofthe vm

Size

Inside the resource block

Vnet name

Cidr block

Nameofthevm

Size

All this are what argument against your resources.

Output

Monday, September 21, 2026 2:28 PM

Once you create the resource using terraform. You will go to the console and verify the resources is created or not.

A vm is launched with your application. To access the same either we need the public ip or the dns name. or lb ip.

I have not given access to azure console.

Public ip is importat.

Output you can use the to display the required content in terraform console or the command line.

Local: when you want to concatenate or want to use name expression.

Calling block: anything I have created outside terraform and I want to include in my terraform script I can use this calling block

Data: I want to launch an windows machine authentication in windows vm happen using password. I will store the password in azure vault. Which is being created outside terraform. While creating windows mahcine we will call this vault to get the password module

Resource group

Monday, September 21, 2026 2:37 PM

Using terraform we will create resource group.
We will use visual studio code editor.
And get the code from terraform registry.

```
Resource azure_rm_resource_group "my-rg" {  
  1.  
  Name : gopal-rg #argument  
  Location : east-us  
}
```

1. Using the reference block terraform store the information inside the state file
2. Reference block is user define using the same we manager all your resources inside your state file.
3. Reference block is also used to map one resource with another resources.

I want to create a vnet

Reference

1. It store the information inside your state file
2. Use it to map other related resources.

Recap

Tuesday, September 22, 2026 9:40 AM

Variables

Tuesday, September 22, 2026 9:57 AM

Variables: what is the use of variables.

1. Store information in variables.
2. Declare it and reuse it

Similar to that once I build an template I need to re use it

Inside walmart every month they hire 30 to 50 fresher.

We need to provide access to azure platform

#there are more than 10k app

##256 department.

#we need to keep on adding this student manually in azure entra id.

We have created a terraform script. For all this 256 deprtement with authentication and authorization rbac.

1. Whenever a new person we just pass the name of the person do terraform apply.

Input variables

Terraform init.

It read all your tf file find out the provider block and create the resource

Terraform plan.

It read your all manifest compare the same with cvloud give you the planner. The order is arranged automatically by terraform.

Rg-gopal

Sap-dev-rg-gopal

#we need to concatenate variables.

#i want to provide multiple tags

Tags: Name

Against the tag we can have multiple expression.

Expressiong

Named expression.

Gopal

Happiness = raised cheeks

Sadness = dropping eyelids

Fear = wide-open eyes

Anger = fixed stare

Against name gopal expression is in key value format

When I call gopal.

Every resource block can create a single resource.

Output block

Tuesday, September 22, 2026

11:39 AM

Vm

Tuesday, September 22, 2026 11:54 AM

Before we create a vm

1. Public ip
2. Nsg: network security port 22 80 443
3. Nic card
4. Vm

Skus: basic it will create the public ip in single az. In the region. One single data center.
Standard it will spread the public ip in multiple az inside your region.

We will create the network security group and attach the nsg with whom. Vm or subnet.

Suppose you have a list of item and you want to apply the list of item inside a single resource group.

Resource group will create only individual resources.

Replicate the same resource group with different value.
Loop for loop

For loop but they called it for_each as meta argument

Nic card

Tuesday, September 22, 2026 12:56 PM

Network interface card

1. Two things we need to attach with our nic
 - a. Private ip : subnet
 - b. Public ip

We can create the virtual machine.

Before we create a virtual machine

1. We need an startup script which will install apache or nginx server. When I put public ip in the browser it should show me some test page.
2. We need to also generate key pair using the same I can login inside the instance also

Recap

Wednesday, September 23, 2026 9:43 AM

We created the virtual machine

We learn about `for_each` loop and how to call those object using `each.key` and `each.value`

Input variables: how to make my code reusable.

There are scenarion where we want to concatinate some values together or I want to use named expression

Local block: it is like

Named expression: tags

Ha

Wednesday, September 23, 2026 10:12 AM

Today we want to create ha- high availability architecture. Creating a resilient architecture

1. Multiple instance deployed in multiple availability zone.
2. Location

I want to deploy my vm in two different az. With different size also

1. Vm in multiple az
2. Different size

We use a variables type will be object.

It is kind of an database

Student database instance db

For the student base what are the default key

Student1 will have key value web1

age = 21 vm_size d1

name=varun az=1

Stream=art subnetid=web

Student2 will have key value

age = 22

name=tom

Stream=commerce

Student3 will have key value

age = 20

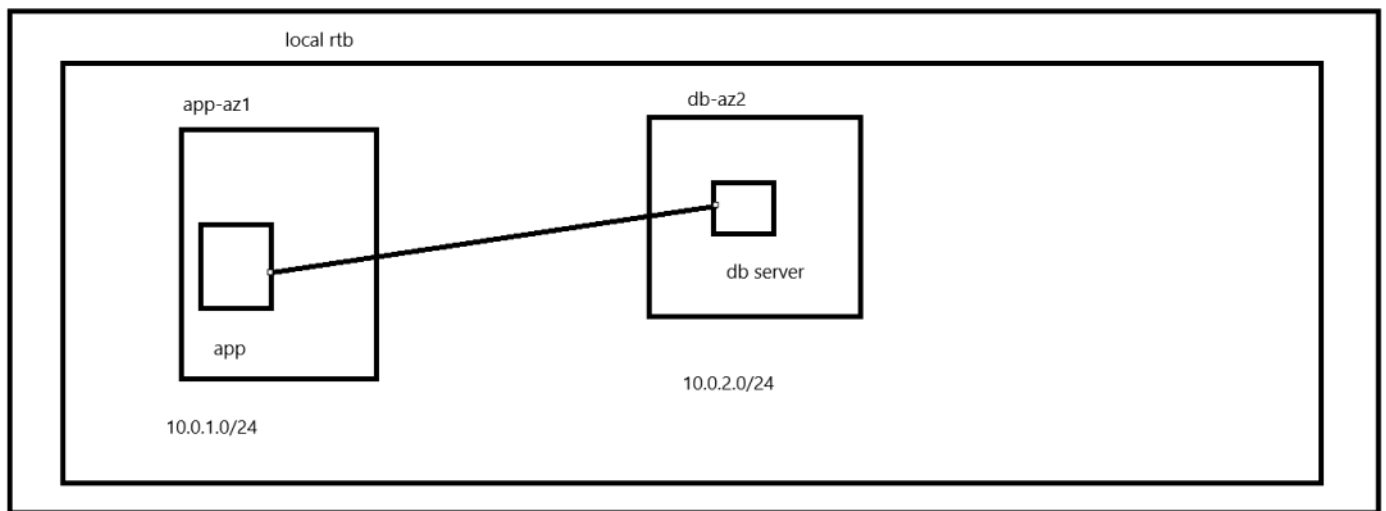
name=bob

Stream=arts

But when I call student db it will have all the three student with there attributes

Select * from student

vnet
10.0.1.0/16 65565



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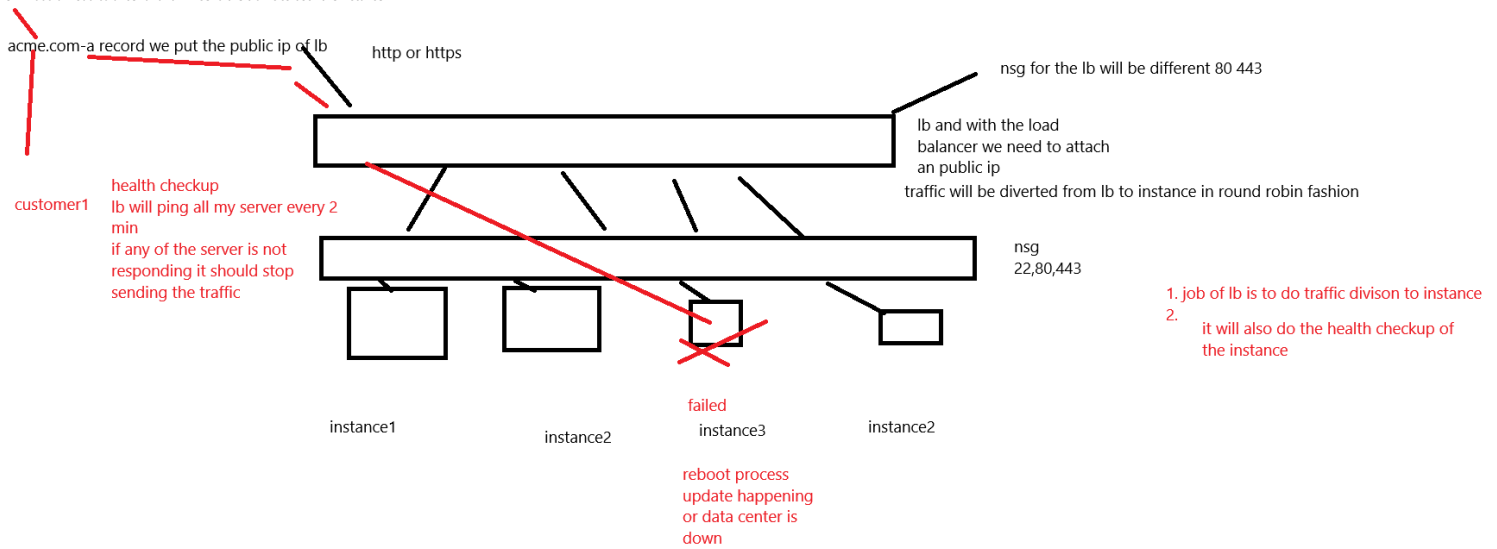
So we need to loop around this variables using for_each

We are going to replace the same with nic card and vm also

Lb

Wednesday, September 23, 2026 12:30 PM

To access my instance using individual public ip I am access the same. Think that this server are like application server
Both will be accessible using single ip address.
#how to do the same.
We will add an load balancer and lb will send the traffic to both the instance



Screen clipping taken: 9/23/2026 12:39 PM

When traffic is coming to the lb

1. We don't need an public ip in the instances.
2. We need to disable the whole public ip block
3. Also from nic card we need to disable the public ip reference

lb

Wednesday, September 23, 2026 2:16 PM

We need to understand the workflow in azure how lb is created according to that we need to write the terraform coding

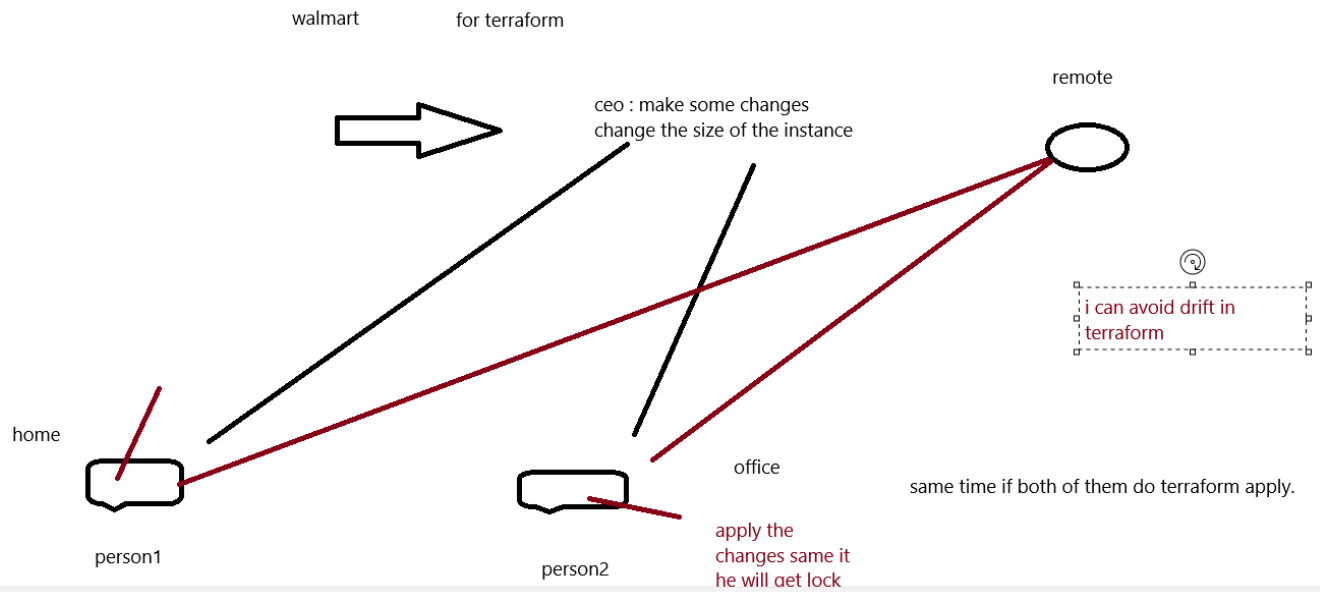
1. To create the lb we need a resource group
2. Name of lb
3. Public facing
4. Front end ip configuration public ip I need to create and attach the same with lb
5. Backend pool: all the vm nic card will be attached to this pool we are clubbing all out instances.
Nic card
6. Inbound rule we need to create from the lb how the traffic will move to backend pool
7. Health checkup

Remote state

Wednesday, September 23, 2026 3:41 PM

Currently we are storing the state file in local machine.

1. When we do terraform apply it will store the information in terraform.tfstate
2. When we do terraform destroy it will take the content of state from terraform.tfstate to terraform.tfstate.backup
3. Vice versa it is happening
 - a. We loosing our previous state.
 - b. Audit team comes they want to audit infrastructure. Never have the previous state.
 - c. We work on collaborative way.



Screen clipping taken: 9/23/2026 4:19 PM

1. We will create the storage account outside terraform.

Git

Thursday, September 24, 2026 9:46 AM

Version control system

1. Git
2. Git global variables
3. Git workflow
4. Git branching
5. Terraform cloud integration

IAC: this is iac so you need to manage the code using a version control system.

Because we are keep on making changes in the code. Want to keep track what changes has been done.

When you want to do ci cd. Continues integration and continues deployment using tools like terraform

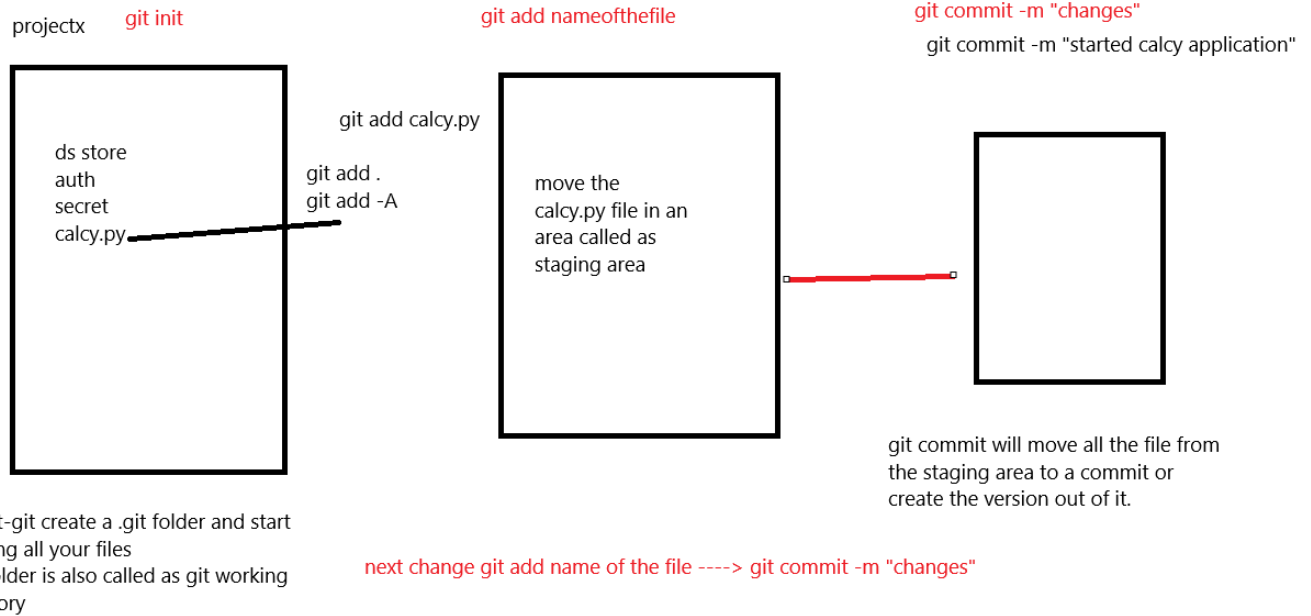
cloud jenkins github action or you can use azure devops.

All this cicd tool do not has versioing. You need to use git

Git is an distributed version control system. All the power of versioing available locally.

Svn: any change I need to do I have to connect the central server.

When you want to collaborate then only connected



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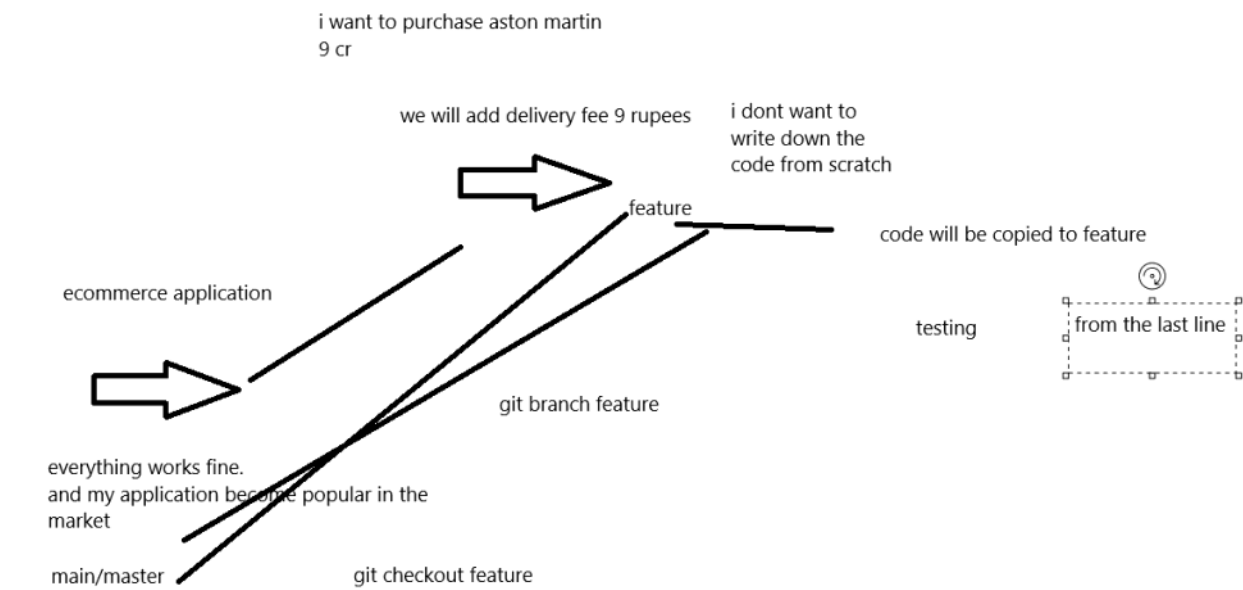
I have create a version in my local system. Change request in object variable change all the instance size.

You need to add the same commit - you need to upload the same in a remote repo. Github gitlab azurerepo bitbucket. They are remote repo or I can say centralized repo for keeping your change or to do the collaboration.

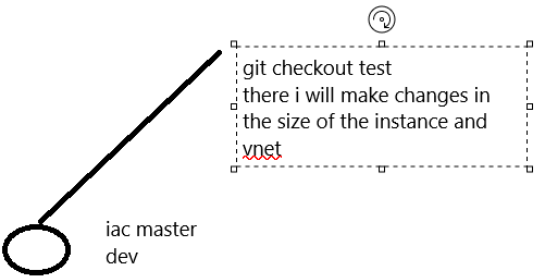
Git branching

Thursday, September 24, 2026 11:48 AM

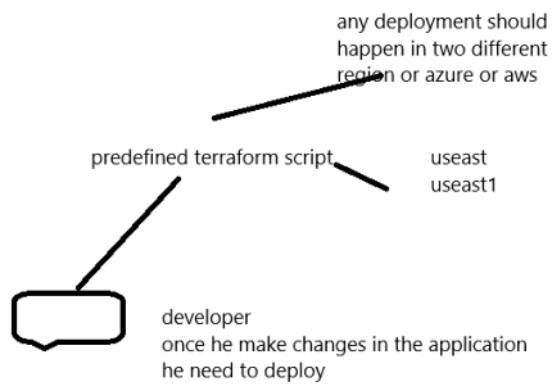
What is the need of branches



Screen clipping taken: 9/24/2026 11:58 AM



Screen clipping taken: 9/24/2026 11:59 AM



Screen clipping taken: 9/24/2026 12:02 PM

Thursday, September 24, 2026 2:17 PM



Git and terraform cloud

Thursday, September 24, 2026 4:24 PM

1. Any change in the terraform code will be available in github
2. Any change in resource creation will be available in terraform cloud.

Soc generale

#once we created the resource

1. destroy

Left

Thursday, September 24, 2026 4:43 PM

Dev test staging from a single code base terraform workspace
Modules in terraform

reusability

Friday, September 25, 2026 9:44 AM

How to achieve reusability.

1. Using terraform cloud and version control system
2. Zero touch deployment.

Once you create the template you can pass a variables file and create multiple environment from the same code base.

Using terraform module we can do the same.

1. We have variables inside the variables we have default value====>terraform apply it will take the default value and create the resource.
2. We created a dev.tfvars I want to overwrite the default value ===== terraform apply --var-file=dev.tfvars

Modules

Friday, September 25, 2026 11:43 AM

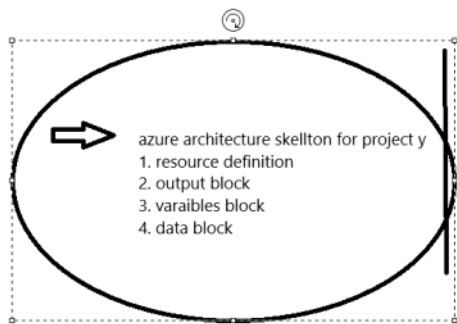
As the name suggest modules.

We are third part vendor for an organization they have organization norms complaine to deploy application in the cloud.

Some of the organization they don't want to write anything from the scratch. Architecture complaine or norm hipaa pci dss and gdpr

They created the sekleton of the application or architecture. They say you can put your values and use the skelton to create resources.

They will upload the same in github.



main.tf

```
source="GitHub.com/gopal1409/test/
project_name = "kyn-project"
environment = "dev"
resource_group_name = "rg-
gopal"
resource_group_location =
"eastus"
vnet_address_space =
["172.31.0.0/16"]
web_subnet_address =
["172.31.1.0/24"]
instance = {
  web1 = {
    vm_size = "Standard_F2as_v7"
    availability_zone = "1"
#each.value.availability_zone 1
    subnet_id = "web"
  }
}
```

- 1. no need to write anything from scratch
- 2. we are following organization standard

terraform init
which will download the code from GitHub
terraform apply

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How to add your code inside the modules folder
From the local machine how to call this module which will create real infrastructure.

- 1. Networking
 - 2. Sg or nsg
 - 3. Compute
 - 4. Lb
- Networking